REFINED QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*SCADA-i - Team DEEP NaN*
Team Repo URL	*https://github.com/deepnan-umhack/scada-i-umhack26*
Project Board URL	*https://safrulbusiness.atlassian.net/jira/software/projects/DEEP/boards/34*
Live Deployment URL	*Dashboard:* *https://scada-i.netlify.app/* *Mobile App:* *https://deep-nan.netlify.app/*

Objective:

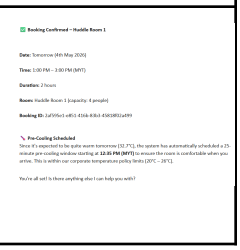
The primary objective is to ensure that SCADA-i can autonomously monitor, analyze, and orchestrate smart building operations by processing real-time IoT data, delegating tasks to specialized AI agents, and executing actions such as energy optimization, workspace booking, and ESG reporting reliably, while maintaining system safety, scalability, and efficient handling of concurrent user requests through robust orchestration, guardrails, and CI/CD checkpoints.

FINAL ROUND (Execution, RCA & Hard Evidence)

No evidence = No points.

1. Test Execution & Evidence (Manual)

*Execute the tests drafted in **Part 1**. You must provide a link to visual proof for at least 3 critical tests.*

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
TC-01	***Happy Case (Entire Flow):*** *End-to-End Flow: Natural*	*Verify that the system can process a natural language request, perform room booking,*	*The system should correctly interpret the user's natural*	*Successfully, Booking Confirmed. The system scheduled*	*Passed*	

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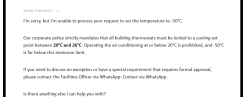
Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
	Language Orchestration + Booking + HVAC Optimization + ESG	*trigger pre-cooling, optimize HVAC based on occupancy, and generate accurate ESG data.*	*language input and successfully complete the booking process. It should automatically trigger pre-cooling before the scheduled time and adjust HVAC settings dynamically based on real-time occupancy data. Additionally, the system should accurately record energy usage and generate ESG data without errors or inconsistencies.*	*Huddle Room 1 for tomorrow from 1:00 PM to 3:00 PM (MYT) with Booking ID 2af595e1-e851-416b-83b3-45818f02a499. Pre-cooling was automatically arranged starting at 12:35 PM based on expected weather conditions, ensuring compliance with corporate temperature policy limits and room comfort upon arrival.*		
TC-02	***Specific Case (Negative):*** *Guardrail Node + AI-driven HVAC Safety*	*Verify that the system's Guardrail Node correctly intercepts and blocks extreme or unsafe HVAC*	*The system should detect the extreme temperature request as invalid and*	*I'm sorry, but I'm unable to process your request to set the temperature to -50°C.*	*Passed*	 SHOW THINKING 1.1 I'm sorry, but I'm unable to process your request to set the temperature to -50°C. Our corporate policy strictly mandates that all building thermostats must be locked to a cooling set-point between **20°C and 26°C**. Operating the air conditioning at or below 20°C is prohibited, and -50°C is far below this minimum limit. If you need to discuss an exception or have a special requirement that requires formal approval, please contact the Facilities Officer via Whatsapp: Contact via Whatsapp Is there anything else I can help you with?

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Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
	+ ESG Integrity	*input values (e.g., -50°C) before they are processed by downstream agents, preventing invalid actions and ensuring system safety.*	*block the payload at the Guardrail Node. It should prevent any HVAC execution, return a clear validation or rejection message to the user, and ensure that no unsafe or non-compliant values are passed to the HVAC system. Additionally, the system should log or flag the incident for monitoring without affecting ESG data integrity.*	*Our corporate policy strictly mandates that all building thermostats must be locked to a cooling set-point between 20°C and 26°C. Operating the air conditioning at or below 20°C is prohibited, and -50°C is far below this minimum limit. If you need to discuss an exception or have a special requirement that requires formal approval, please contact the Facilities Officer via WhatsApp: Contact via*		

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Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
				WhatsApp Is there anything else I can help you with?		
TC-03	***NFR (Performance):*** *Natural Language Booking + Orchestrator Load Handling*	*Verify that the system can handle multiple concurrent booking requests through the chat interface.*	*The system should process all booking requests efficiently without system failure or significant delay. It should correctly assign rooms without duplication and maintain a response time within acceptable limits, ensuring stable system performance under load.*	*Average Response time is received as 20s.* *0% Error Rate.*	*Passed*	*agents/test_regression.py::test_chat_endpoint[Booking-I need a room for 10 people tomorrow at 2 PM for 1 hour.] [Booking] Response time: 24.53s [Booking] Agent reply: I found several rooms available for **tomorrow (Monday, 4 May 2026) at 2:00 PM MYT** for **1 hour** that can accommodate **10 people**: 1. **Room 204 (Boardroom)** – Capacity: 10 (Features: Projector, Whiteboard, AC) 2. **Syndicate Room** – Capacity: 20 3. **Biology Lab 1** – Capacity: 24 (Features: Lab Equipment, Smart*

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Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
						*AC, Projector) 4. **Biology Lab 2** – Capacity: 24 (Features: Lab Equipment, Smart AC, Projector) 5. **Computer Lab 1** – Capacity: 35 (Features: Computers, Whiteboard, Smart AC) Which room would you like to book? PASSED agents/test_regression.py::test_chat_endpoint[HVAC-Set the temperature in Huddle Room 1 to 22 degrees Celsius.] [HVAC] Response time: 33.03s [HVAC] Agent reply: The temperature in **Huddle Room 1** has been successfully set to **22 degrees Celsius** as requested. The air conditioner is now running in **Cool mode** at **high fan*

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Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
						*speed** to bring the room to your desired temperature. Is there anything else I can help you with? PASSED agents/test_regression.py::test_chat_endpoint[ESG-Is setting the AC to 18 degrees compliant with ESG policy?] [ESG] Response time: 3.87s [ESG] Agent reply: Access Denied: You do not have the required administrative permissions for this action. PASSED ========== 3 passed in 62.10s (0:01:02) ==========*
TC- 04	***Security (Access Control):*** *Role-Based Authorization + Agent Permission Enforcement*	*Verify that the system correctly restricts unauthorized users from performing privileged actions (e.g., accessing ESG reports or*	*The system should deny access to the requested action.* *A clear message should be returned*	*Access Denied: You do not have the required administrative permissions for this action.*	*Passed*	SHOW THOUGHTS ^ Inform the user they are denied access as they do not have the required permissions to view ESG reports. Access Denied: You do not have the required administrative permissions for this action.

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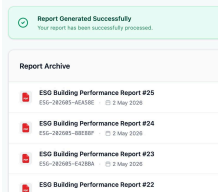
Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
		modifying HVAC settings in restricted rooms).	*(e.g., "Access Denied: insufficient permissions").* *No backend action (HVAC / ESG report generation) should be executed.* *The system should maintain normal operation without errors.*			
TC-05	***NFR (Performance): ESG Agent + Data Integrity***	*Verify that ESG reports are generated accurately based on validated system data.*	*The system should generate ESG reports that accurately reflect real energy consumption and carbon reduction based on verified sensor and system data. The report should not include any fabricated or AI-generated estimations, ensuring full*	*The ESG agent successfully generated ESG reports that accurately matched the raw telemetry data.*	*Passed*	

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Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
			data integrity and reliability.			

2. Automated Testing Pipeline

2.1. Unit Testing (Core Logic & Functions)

Unit testing is primarily focused on Isolation of the individual components or functions to ensure that these functionalities work perfectly on their own without the need of any external dependencies. Mock your databases or APIs here.

- ***Test Runner Used:*** *Vitest + React Testing Library*
- ***Targeted Components:*** *BrowseSpaces*
- ***File(s) Covered:*** *BrowseSpaces_unittest.test.tsx*

Unit Test ID	*Function/Component Tested*	*Test Scenario*	*Expected Outcome*	*Actual Result*	*Status*
UT-01	*BrowseSpaces render*	*Render component with default props*	*"Spaces List" heading is displayed*	*renders core heading 34ms.*	***Passed***
UT-02	*onSpaceSelected callback*	*User clicks "Syndicate Room" card*	*onSpaceSelected('Syndicate Room') is called*	*calls onSpaceSelected with selected room name 10ms*	***Passed***

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UT-03	*onBack callback*	*User clicks close button (✕)*	*onBack is called once interaction occurs*	*calls onBack when close button is clicked 7ms*	***Passed***
UT-04	*onOpenBookingStatus callback*	*User clicks "Bookings" button*	*onOpenBookingStatus is called*	*callsonOpenBookingStatus when Bookings is clicked 7ms*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\website>npx vitest run src/pages/unit_tests/BrowseSpaces_unittest.test.tsx

 RUN  v4.1.5 D:/Hackathons/scada-i-umhack26-fork/website

 ✓ src/pages/unit_tests/BrowseSpaces_unittest.test.tsx (4 tests) 61ms
   ✓ BrowseSpaces (targeted unit tests) (4)
     ✓ UT-01: renders core heading 34ms
     ✓ UT-02: calls onSpaceSelected with selected room name 10ms
     ✓ UT-03: calls onBack when close button is clicked 7ms
     ✓ UT-04: calls onOpenBookingStatus when Bookings is clicked 7ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  12:22:39
   Duration  2.26s (transform 187ms, setup 84ms, import 1.41s, tests 61ms, environment 566ms)
```

- ***Test Runner Used:*** *Vitest + React Testing Library*
- ***Targeted Components:*** *LandingPage*
- ***File(s) Covered:*** *LandingPage_unittest.test.tsx*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*LandingPage logo render*	*Render LandingPage and query by alt text "DeepNaN Logo"*	*Logo image should be present in DOM*	*renders DeepNaN logo 20ms*	***Passed***

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UT-02	*LandingPage logo source binding*	*Render page and verify logo src*	*Logo src should match mocked value "logo.svg"*	*uses mocked logo src 2ms*	***Passed***
UT-03	*LandingPage splash ripple rendering*	*Render page and count ripple elements using .border-slate-400 selector*	*Exactly 2 splash ripple elements should render*	*renders exactly two splash ripple elements 3ms*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\website>npx vitest run src/pages/unit_tests/LandingPage_unittest.test.tsx

 RUN  v4.1.5 D:/Hackathons/scada-i-umhack26-fork/website

 ✓ src/pages/unit_tests/LandingPage_unittest.test.tsx (3 tests) 29ms
   ✓ LandingPage (targeted unit tests) (3)
     ✓ UT-01: renders DeepNaN logo 20ms
     ✓ UT-02: uses mocked logo src 2ms
     ✓ UT-03: renders exactly two splash ripple elements 3ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  12:46:33
   Duration  1.02s (transform 66ms, setup 83ms, import 170ms, tests 29ms, environment 589ms)
```

- ***Test Runner Used:*** *Vitest + React Testing Library*
- ***Targeted Components:*** *AuthPage*
- ***File(s) Covered:*** *AuthPage_unittest.test.tsx*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*AuthPage default login UI*	*Render page with onLoginSuccess prop*	*Email + Password input fields are shown*	*renders login form fields by default 22ms*	***Passed***

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UT-02	*Login success flow*	*Mock signInWithPassword to return { error: null }, submit form*	*onLoginSuccess callback is called*	*calls onLoginSuccess when login succeeds 16ms*	***Passed***
UT-03	*Login failure flow*	*Mock signInWithPassword to return { error: { message: 'Invalid credentials' } }, submit form*	*Alert shown: "Login failed: Invalid credentials"*	*shows alert when login fails 11ms*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\website>npx vitest run src/pages/unit_tests/AuthPage_unittest.test.tsx

 RUN  v4.1.5 D:/Hackathons/scada-i-umhack26-fork/website

 ✓ src/pages/unit_tests/AuthPage_unittest.test.tsx (3 tests) 141ms
   ✓ AuthPage (targeted unit tests) (3)
     ✓ UT-01: renders login form fields by default 25ms
     ✓ UT-02: calls onLoginSuccess when login succeeds 81ms
     ✓ UT-03: shows alert when login fails 32ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  22:07:34
   Duration  1.09s (transform 49ms, setup 79ms, import 139ms, tests 141ms, environment 590ms)
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *_parse_target_time from hvac_toolkit.py*
- ***File(s) Covered:*** *parse_target_time_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*_parse_target_time*	*Input datetime with trailing Z (2026-04-24T14:00:00Z)*	*Parsed as UTC-aware datetime (+00:00)*	*Recorded as 2026-04-24T14:00:00+00:00*	***Passed***

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UT-02	*_parse_target_time*	*Input datetime with explicit offset (+08:00)*	*Offset preserved in parsed datetime*	*Recorded as 2026-04-24T14:00:00+08:00*	***Passed***
UT-03	*_parse_target_time*	*Input datetime with extra surrounding whitespace*	*Whitespace trimmed and datetime parsed correctly*	*Recorded as 2026-04-24T14:00:00+00:00*	***Passed***
UT-04a	*_parse_target_time*	*Invalid input: empty string*	*Raises ValueError*	*Recorded: ValueError raised as expected*	***Passed***
UT-04b	*_parse_target_time*	*Invalid input: not-a-date*	*Raises ValueError*	*Recorded: ValueError raised as expected*	***Passed***
UT-04c	*_parse_target_time*	*Invalid input: 2026/04/24 14:00*	*Raises ValueError*	*Recorded: ValueError raised as expected*	***Passed***

Execution Proof:

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```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest hvac_agent/tests/unit_tests/parse_target_time_unitt
est.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 7 items

hvac_agent\tests\unit_tests\parse_target_time_unittest.py ......UT-01 | Trailing Z UTC parse | Actual Result: 2
026-04-24T14:00:00+00:00 | Pass
UT-02 | Offset datetime parse | Actual Result: 2026-04-24T14:00:00+08:00 | Pass
UT-03 | Whitespace-trimmed parse | Actual Result: 2026-04-24T14:00:00+00:00 | Pass
UT-04 | Invalid input '' | Actual Result: ValueError raised as expected | Pass
UT-04 | Invalid input 'not-a-date' | Actual Result: ValueError raised as expected | Pass
UT-04 | Invalid input '2026/04/24 14:00' | Actual Result: ValueError raised as expected | Pass
.

========================================== 7 passed in 0.10s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *_select_nearest_hour_index from hvac_toolkit.py*
- ***File(s) Covered:*** *select_nearest_hour_index_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*_select_nearest_hour_index*	*Exact match scenario (target=10:00:45) against hourly list [09:00,10:00, 11:00]*	*Returns index 1*	*Returned index 1*	***Passed***
UT-02	*_select_nearest_hour_index*	*No exact minute match (target=10:35), function hour-snaps to 10:00*	*Returns index 1*	*Returned index 1*	***Passed***
UT-03	*_select_nearest_hour_index*	*Timezone-aware UTC target (13:20+00:00) converted to KL hour (21:00)*	*Returns index 1 for 21:00 in [20:00,21:00,22:00]*	*Returned index 1*	***Passed***

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UT-04a	*_select_nearest_hour_index*	*Param case target=09:29 (hour-snap behavior set)*	*Returns index 0*	*Returned index 0*	***Passed***
UT-04b	*_select_nearest_hour_index*	*Param case target=09:31 (hour-snap behavior set)*	*Returns index 0*	*Returned index 0*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest hvac_agent/tests/unit_tests/select_nearest_hour_ind
ex_unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 6 items

hvac_agent\tests\unit_tests\select_nearest_hour_index_unittest.py .....UT-01 | Exact hour match | Actual Result
: Returned index 1 | Pass
UT-02 | Hour-snapped match for 10:35 | Actual Result: Returned index 1 | Pass
UT-03 | Timezone-aware target converted to KL hour | Actual Result: Returned index 1 | Pass
UT-04 | Hour snap nearest case for 2026-04-24T09:29:00 | Actual Result: Returned index 0 | Pass
UT-04 | Hour snap nearest case for 2026-04-24T09:31:00 | Actual Result: Returned index 0 | Pass
.

========================================== 6 passed in 0.15s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *calculate_weather_aware_precool_start from hvac_toolkit*
- ***File(s) Covered:*** *calculate_weather_aware_precool_start_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*calculate_weather_aware_precool_start*	*Invalid booking_start input ("not-a-date")*	*Returns failure response with pre_cool_start=None and invalid-form*	*status=failed, pre_cool_start=None*	***Passed***

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			at reason		
UT-02	*calculate_weather_aware_precool_start*	*Weather service unavailable (status=failed)*	*Falls back to default 20-minute pre-cool window and computes pre_cool_start = booking_start - 20 min*	*pre_cool_start=2026-04-24T13:40:00+08:00, duration=20*	***Passed***
UT-03	*calculate_weather_aware_precool_start*	*Extreme heat band (outside_temperature_c=35.0)*	*Uses 30-minute recommendation*	*outside_temperature_c=35.0, duration=30*	***Passed***
UT-04a	*calculate_weather_aware_precool_start*	*Temperature band case 32.0°C*	*Uses 25-minute recommendation*	*duration=25*	***Passed***
UT-04b	*calculate_weather_aware_precool_start*	*Temperature band case 28.0°C*	*Uses 20-minute recommendation*	*duration=20*	***Passed***

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UT-04c	*calculate_weather_aware_precool_start*	*Temperature band case 27.9°C*	*Uses 15-minute recommendation*	*duration=15*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest hvac_agent/tests/unit_tests/calculate_weather_aware
_precool_start_unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 5 items

hvac_agent\tests\unit_tests\calculate_weather_aware_precool_start_unittest.py 🔧Tool called: calculate_weather_
aware_precool_start
📅 booking_start: not-a-date
🏢 room_id: Room
.🔧Tool called: calculate_weather_aware_precool_start
📅 booking_start: 2026-04-24T14:00:00+08:00
🏢 room_id: Room
.🔧Tool called: calculate_weather_aware_precool_start
📅 booking_start: 2026-04-24T14:00:00+08:00
🏢 room_id: Room
.🔧Tool called: calculate_weather_aware_precool_start
📅 booking_start: 2026-04-24T14:00:00+08:00
🏢 room_id: Room
🔧Tool called: calculate_weather_aware_precool_start
📅 booking_start: 2026-04-24T14:00:00+08:00
🏢 room_id: Room
🔧Tool called: calculate_weather_aware_precool_start
📅 booking_start: 2026-04-24T14:00:00+08:00
🏢 room_id: Room
.UT-01 | Invalid booking_start | Actual Result: status=failed, pre_cool_start=None | Pass
UT-02 | Weather unavailable fallback | Actual Result: pre_cool_start=2026-04-24T13:40:00+08:00, duration=20 | P
ass
UT-03 | Extreme heat band | Actual Result: outside_temperature_c=35.0, duration=30 | Pass
UT-04 | Temperature band for 32.0C | Actual Result: duration=25 | Pass
UT-04 | Temperature band for 28.0C | Actual Result: duration=20 | Pass
UT-04 | Temperature band for 27.9C | Actual Result: duration=15 | Pass
.

========================================== 5 passed in 0.26s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *run_hvac_request from hvac_toolkit*
- ***File(s) Covered:*** *run_hvac_request_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*run_hvac_request*	*handle_hvac_request succeeds and returns a normal payload*	*Function should return handler payload unchanged*	*action=SET_HVAC, execution=sent*	***Passed***

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UT-02	*run_hvac_request*	*handle_hvac_request raises runtime exception*	*Function should return structured error payload with decision.action="ERROR" and execution.status="failed"*	*action=ERROR, execution=failed, includes error reason*	***Passed***
UT-03	*run_hvac_request*	*Error path triggered during request handling*	*Error payload should still include generated non-empty request_id*	*request_id=<generated uuid string> present*	***Passed***

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```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest hvac_agent/tests/unit_tests/run_hvac_request_unitte
st.py -s
========================================= test session starts =========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

hvac_agent\tests\unit_tests\run_hvac_request_unittest.py 🔧Tool called: run_hvac_request
🆔 request_id: d2a81c7b-6f8f-4c82-a2d9-0dd4792b81c9
👤 user_id: user1
🏢 room_id: RoomA
📝 request_type: SET_TEMPERATURE
🌡️requested_temperature: 24.0
.🔧Tool called: run_hvac_request
🆔 request_id: 9d6afc6d-d936-472b-bcfa-ecbce660c16d
👤 user_id: user1
🏢 room_id: RoomA
📝 request_type: SET_TEMPERATURE
🌡️requested_temperature: 24.0
❌ Tool Execution Error: simulated failure
.🔧Tool called: run_hvac_request
🆔 request_id: 93e6ba54-bf6d-4ec0-bbf8-078292528d7b
👤 user_id: user2
🏢 room_id: RoomB
📝 request_type: NO_USER_REQUEST
❌ Tool Execution Error: boom
.UT-01 | Success passthrough | Actual Result: action=SET_HVAC, execution=sent | Pass
UT-02 | Exception fallback payload | Actual Result: action=ERROR, execution=failed, reason=simulated failure |
Pass
UT-03 | Error payload includes request_id | Actual Result: request_id=93e6ba54-bf6d-4ec0-bbf8-078292528d7b | Pa
ss
.

========================================= 4 passed in 0.24s =========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *check_equipment_availability from tools*
- ***File(s) Covered:*** *check_equipment_availability_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*check_equipment_availability_tool*	*DATABASE_URL is missing (None)*	*Returns an error payload indicating config/database URL issue*	*status=error, message=<error message from payload>*	***Passed***
UT-02	*check_equipment_availability_tool*	*DB connection mocked with filtered inventory rows for ["projector","]*	*Returns success payload and matching inventory entries*	*status=success, items=2*	***Passed***

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		mic"]			

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest booking_agent/tools/unit_tests/check_equipment_avai
lability_unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 3 items

booking_agent\tools\unit_tests\check_equipment_availability_unittest.py ..UT-01 | Missing DATABASE_URL | Actual
 Result: status=error, message=DATABASE_URL is not configured. | Pass
UT-02 | Filtered inventory lookup | Actual Result: status=success, items=2 | Pass
.

========================================== 3 passed in 0.03s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *check_room_availability from tools*
- ***File(s) Covered:*** *check_room_availability_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*check_room_availability_tool*	*DATABASE_URL missing → fallback room list path*	*Returns success with at least one available room*	*status=success, rooms=<count>*	***Passed***
UT-02	*check_room_availability_tool*	*required_features=["projector"] filter*	*Returns success and all returned rooms include projector feature*	*status=success, rooms=<count>*	***Passed***

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UT-03	*check_room_availability_tool*	*required_features=["non-existent-feature"]*	*Returns no-match response*	*status=no_rooms_found, message=<payload message>*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest booking_agent/tools/unit_tests/check_room_availabil
ity_unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

booking_agent\tools\unit_tests\check_room_availability_unittest.py ...UT-01 | Fallback room list | Actual Resul
t: status=success, rooms=2 | Pass
UT-02 | Required features filter | Actual Result: status=success, rooms=1 | Pass
UT-03 | No matching features | Actual Result: status=no_rooms_found, message=No rooms available for this time a
nd capacity. | Pass
.

========================================== 4 passed in 0.05s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *create_booking_tool from tools*
- ***File(s) Covered:*** *create_booking_record_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*create_booking_tool*	*DATABASE_URL is missing*	*Returns error payload indicating DB URL not configured*	*status=error, message=DATABASE_URL is not configured.*	***Passed***
UT-02	*create_booking_tool*	*Invalid target_department_ids UUID string (not-a-uuid)*	*Returns validation-style error payload for invalid department*	*status=error, message=Invalid department UUID: not-a-uuid*	***Passed***

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			UUID		
UT-03	*create_booking_tool*	*Invalid target_department_ids UUID string (not-a-uuid)*	*Returns success payload with booking ID and normalized department IDs*	*status=success, booking_id=22222222-2222-2222-2222-222222222222, departments=1*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest booking_agent/tools/unit_tests/create_booking_recor
d_unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

booking_agent\tools\unit_tests\create_booking_record_unittest.py ...UT-01 | Missing DATABASE_URL | Actual Resul
t: status=error, message=DATABASE_URL is not configured. | Pass
UT-02 | Invalid department UUID | Actual Result: status=error, message=Invalid department UUID: not-a-uuid | Pa
ss
UT-03 | Valid booking payload | Actual Result: status=success, booking_id=22222222-2222-2222-2222-222222222222,
 departments=1 | Pass
.

========================================== 4 passed in 0.03s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *get_room_directory_tool from tools*
- ***File(s) Covered:*** *get_room_directory_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*get_room_directory_tool*	*DATABASE_URL missing*	*Returns error payload for missing DB config*	*status=error, message=DATABASE_URL is not configured.*	***Passed***

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UT-02	*get_room_directory_tool*	*Query returns no rows*	*Returns no_rooms response with no-match message*	*status=no_rooms, message=No room types match this criteria.*	***Passed***
UT-03	*get_room_directory_tool*	*Rows returned; filter by required_features=["projector"]; aggregate by room type*	*Returns success payload with filtered/aggregated room-type summary*	*status=success, room_types=1, top_type=boardroom*	***Passed***

Execution Proof:

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest booking_agent/tools/unit_tests/get_room_directory_u
nittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

booking_agent\tools\unit_tests\get_room_directory_unittest.py ...UT-01 | Missing DATABASE_URL | Actual Result:
status=error, message=DATABASE_URL is not configured. | Pass
UT-02 | Empty room query result | Actual Result: status=no_rooms, message=No room types match this criteria. |
Pass
UT-03 | Aggregated types with required feature filter | Actual Result: status=success, room_types=1, top_type=b
oardroom | Pass
.

========================================== 4 passed in 0.03s ==========================================
```

- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *generate_esg_report_tool from tools*
- ***File(s) Covered:*** *generate_esg_report_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***

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UT-01	*generate_esg_report_tool*	*Invalid ISO date input (start_date="not-a-date")*	*Returns date-format error string and stops*	*result=ERROR: Invalid date format. Please use ISO 8601. DO NOT RETRY.*	***Passed***
UT-02	*generate_esg_report_tool*	*Missing environment config (DATABASE_URL=None, SUPABASE_URL=None)*	*Returns critical configuration error string*	*result=CRITICAL ERROR: Database or Supabase URL is not configured. DO NOT RETRY.*	***Passed***
UT-03	*generate_esg_report_tool*	*Successful flow with mocked asyncio.to_thread + mocked DB connect*	*Returns JSON success payload with status=success and PDF URL*	*status=success, report_id=<generated ESG-YYYYMM-XXXXXX>*	***Passed***

Execution Proof:

```
========================================== 4 passed in 0.03s ==========================================

D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest esg_agent/tools/unit_tests/generate_esg_report_unit
test.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

esg_agent\tools\unit_tests\generate_esg_report_unittest.py ...UT-01 | Invalid ISO date input | Actual Result: r
esult=ERROR: Invalid date format. Please use ISO 8601. DO NOT RETRY. | Pass
UT-02 | Missing DB/Supabase config | Actual Result: result=CRITICAL ERROR: Database or Supabase URL is not conf
igured. DO NOT RETRY. | Pass
UT-03 | Successful ESG report flow | Actual Result: status=success, report_id=ESG-202605-7AED16 | Pass
.

========================================== 4 passed in 0.04s ==========================================
```

- ***Test Runner Used:*** *Pytest*

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- ***Targeted Components:*** *search_esg_policy_tool from tools*
- ***File(s) Covered:*** *policy_retriever_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*search_esg_policy_tool*	*No matched policy documents from retriever*	*Returns "no policy rules found" error string*	*result=ERROR: No specific policy rules found for this query. DO NOT RETRY.*	***Passed***
UT-02	*search_esg_policy_tool*	*Matched documents returned from mocked RPC*	*Returns JSON success payload with formatted results list*	*status=success, results=2*	***Passed***
UT-03	*search_esg_policy_tool*	*Dependency failure (_build_clients raises exception)*	*Returns critical error string*	*result=CRITICAL ERROR: embedding service down*	***Passed***

Execution Proof*:*

```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest esg_agent/tools/unit_tests/policy_retriever_unittes
t.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

esg_agent\tools\unit_tests\policy_retriever_unittest.py ...UT-01 | No matching policy docs | Actual Result: res
ult=ERROR: No specific policy rules found for this query. DO NOT RETRY. | Pass
UT-02 | Policy retrieval success | Actual Result: status=success, results=2 | Pass
UT-03 | Dependency exception path | Actual Result: result=CRITICAL ERROR: embedding service down | Pass
.

========================================== 4 passed in 0.03s ==========================================
```

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- ***Test Runner Used:*** *Pytest*
- ***Targeted Components:*** *analyze_hvac_compliance_tool from tools*
- ***File(s) Covered:*** *analyze_hvac_compliance_unittest.py*

Unit Test ID	***Function/Component Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
UT-01	*analyze_hvac_compliance_tool*	*Invalid date string for start_date ("bad-date")*	*Returns date-format validation error message*	*result=ERROR: Invalid date format. Please use ISO 8601 (e.g., YYYY-MM-DDTHH:MM:SSZ). DO NOT RETRY.*	***Passed***
UT-02	*analyze_hvac_compliance_tool*	*DATABASE_URL missing (None)*	*Returns critical configuration error for missing DB URL*	*result=CRITICAL ERROR: DATABASE_URL is not configured. DO NOT RETRY.*	***Passed***
UT-03	*analyze_hvac_compliance_tool*	*Success path with mocked DB rows (mixed occupancy + low temp values)*	*Returns success JSON with computed audit metrics (total, wastage, extreme, score)*	*status=success, total=3, wastage=2, extreme=2, score=0*	***Passed***

Execution Proof*:*

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```
D:\Hackathons\scada-i-umhack26-fork\agents>python -m pytest esg_agent/tools/unit_tests/analyze_hvac_compliance_
unittest.py -s
========================================== test session starts ==========================================
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\Hackathons\scada-i-umhack26-fork
configfile: pytest.ini
plugins: anyio-4.13.0, langsmith-0.7.33, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scop
e=function
collected 4 items

esg_agent\tools\unit_tests\analyze_hvac_compliance_unittest.py ...UT-01 | Invalid date format | Actual Result:
result=ERROR: Invalid date format. Please use ISO 8601 (e.g., YYYY-MM-DDTHH:MM:SSZ). DO NOT RETRY. | Pass
UT-02 | Missing DATABASE_URL | Actual Result: result=CRITICAL ERROR: DATABASE_URL is not configured. DO NOT RET
RY. | Pass
UT-03 | Success with mixed occupancy/temperature rows | Actual Result: status=success, total=3, wastage=2, extr
eme=2, score=0 | Pass
.

========================================== 4 passed in 0.03s ==========================================
```

2.2. Integration & API Testing (System Modules)

This test is mainly conducting communication (e.g., Frontend to Backend, Backend to Database). Do they integrate correctly?

- **Test Tool Used:** Postman automated collections (Newman CLI via GitHub Actions)
- **Targeted Integrations:** Backend REST API connecting to Supabase PostgreSQL database
- **File(s) Covered:** .github/workflows/postman-test.yml, postman/collections/HVAC Agent API Tests/collection.json

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
IT-01	*GET /rest/v1/departments → Supabase*	*Sends a GET request to fetch all departments from Supabase database*	*Returns HTTP 200 with an array of department records under 2000ms*	*Returns HTTP 200 with array of department data*	***Passed***
IT-02	*POST /rest/v1/departments → Supabase*	*Sends a POST request to insert a new test department with name and description fields*	*Returns HTTP 201 with the created department record containing name field*	*Returns HTTP 201 with inserted department object*	***Passed***

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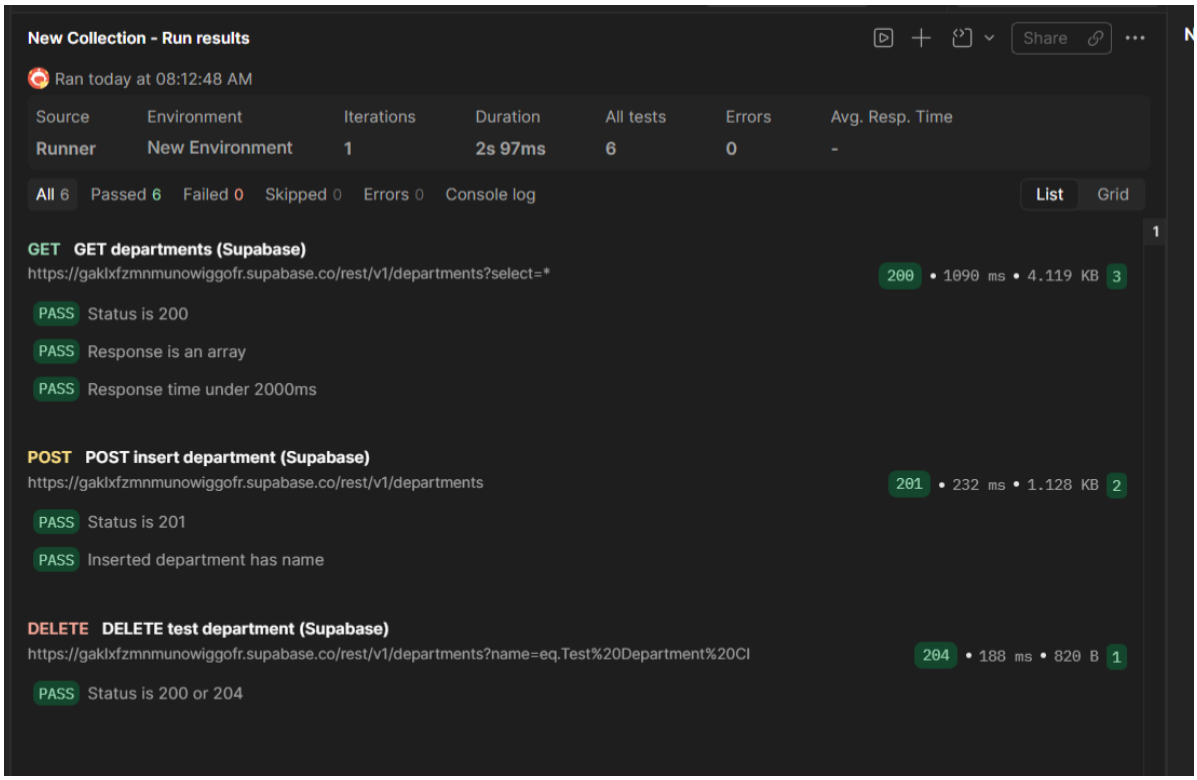
IT-03	*DELETE /rest/v1/departments → Supabase*	*Sends a DELETE request to remove the test department by name filter*	*Returns HTTP 200 or 204 confirming record deletion*	*Returns HTTP 204 No Content*	***Passed***

Execution Proof:



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- **Test Tool Used:** Pytest + requests (Python, via GitHub Actions)
- **Targeted Integrations:** /chat API endpoint — Booking, HVAC, ESG agent flows
- **File(s) Covered:** .github/workflows/regression.yml, agents/test_regression.py

Test ID	***Integration / Point Tested***	***Test Scenario***	***Expected Outcome***	***Actual Result***	***Status***
IT-01	*Booking Agent → /chat*	*Sends a POST request with a room booking message for 10 people tomorrow at 2 PM for 1 hour*	*Returns HTTP 200 with a final_response event containing a non-empty reply in the stream*	*[Booking] Response time: 24.53s* *[Booking] Agent reply: I found several rooms available for **tomorrow (Monday, 4 May 2026) at 2:00 PM MYT** for **1 hour** that can accommodate **10 people**:* *1. **Room 204 (Boardroom)** – Capacity: 10 (Features: Projector, Whiteboard, AC)* *2. **Syndicate Room** – Capacity: 20* *3. **Biology Lab 1** – Capacity: 24 (Features: Lab Equipment, Smart AC, Projector)* *4. **Biology Lab 2** – Capacity: 24 (Features: Lab Equipment, Smart AC, Projector)* *5. **Computer Lab 1** – Capacity: 35 (Features: Computers, Whiteboard, Smart AC)*	***Passed***

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				Which room would you like to book?	
IT-02	*HVAC Agent → /chat*	*Sends a POST request to set the temperature in Huddle Room 1 to 22 degrees Celsius*	*Returns HTTP 200 with a final_response event containing a non-empty reply in the stream*	*[HVAC] Response time: 33.03s* *[HVAC] Agent reply: The temperature in **Huddle Room 1** has been successfully set to **22 degrees Celsius** as requested. The air conditioner is now running in **Cool mode** at **high fan speed** to bring the room to your desired temperature.* *Is there anything else I can help you with?*	***Passed***
IT-03	*ESG Agent → /chat*	*Sends a POST request asking whether setting AC to 18 degrees complies with ESG policy*	*Returns HTTP 200 with a final_response event containing a non-empty reply in the stream*	*[ESG] Response time: 3.87s* *[ESG] Agent reply: Access Denied: You do not have the required administrative permissions for this action.*	***Passed***

Execution Proof:

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```
▶ Run pytest agents/test_regression.py -v -s
============================= test session starts ==============================
platform linux -- Python 3.11.15, pytest-9.0.3, pluggy-1.6.0 -- /opt/hostedtoolcache/Python/3.11.15/x64/bin/python
cachedir: .pytest_cache
rootdir: /home/runner/work/scada-i-umhack26/scada-i-umhack26
configfile: pytest.ini
collecting ... collected 3 items
agents/test_regression.py::test_chat_endpoint[Booking-I need a room for 10 people tomorrow at 2 PM for 1 hour.]
[Booking] Response time: 24.53s
[Booking] Agent reply: I found several rooms available for **tomorrow (Monday, 4 May 2026) at 2:00 PM MYT** for **1 hour** that can accommodate **10 people**:
1. **Room 204 (Boardroom)** – Capacity: 10 (Features: Projector, Whiteboard, AC)
2. **Syndicate Room** – Capacity: 20
3. **Biology Lab 1** – Capacity: 24 (Features: Lab Equipment, Smart AC, Projector)
4. **Biology Lab 2** – Capacity: 24 (Features: Lab Equipment, Smart AC, Projector)
5. **Computer Lab 1** – Capacity: 35 (Features: Computers, Whiteboard, Smart AC)
Which room would you like to book?
PASSED
agents/test_regression.py::test_chat_endpoint[HVAC-Set the temperature in Huddle Room 1 to 22 degrees Celsius.]
[HVAC] Response time: 33.03s
[HVAC] Agent reply: The temperature in **Huddle Room 1** has been successfully set to **22 degrees Celsius** as requested. The air conditioner is now running in **Cool mode** at
**high fan speed** to bring the room to your desired temperature.
Is there anything else I can help you with?
PASSED
agents/test_regression.py::test_chat_endpoint[ESG-Is setting the AC to 18 degrees compliant with ESG policy?]
[ESG] Response time: 3.87s
[ESG] Agent reply: Access Denied: You do not have the required administrative permissions for this action.
PASSED
======================== 3 passed in 62.10s (0:01:02) ========================
```

3. Defect Log & Fix Traceability (Root Cause Analysis)

*Log the bugs you **fixed**. Include the exact Git Commit Hash to prove the fix. AI cannot fake your codebase's commit history.*

Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
DEF-01	*There is a navigation issue when re-selecting tags. While initial tag selection and replies work as expected, clicking an already active tag triggers a page refresh or redirects to the homepage. This causes the current chat session to be lost and initiates a new session with*	*1. Navigate to the chat interface and select an initial set of tags.* *2. Type an input into the chatbox and submit.* *3. Verify the AI generates a response successfully.* *4. While still in the same chat session, attempt to click a different tag.*	No console errors. UI successfully executes navigation to '/' instead of updating local state.	*The bug is logical rather than functional. The tag component triggered an unintended global refresh because it lacked event prevention. Furthermore, the application lacked a persistence mechanism (such as*	*https://github.com/deepnan-umhack/scada-i-umhack26/commit/c17a3082e478b13be16e162a8912c1df943a43ca]*

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Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
	that tag instead of maintaining the existing conversation state.	*5. Observe the application redirecting to the main page with the new tag but the previous chat history disappearing.*		*LocalStorage or SessionStorage) , meaning the chat state was stored only in volatile memory and could not be recovered after the refresh.*	
DEF-02	*The website allows registration using invalid email formats (e.g., "test@invalid") without a proper domain.*	*1. Go to the Sign-Up page.* *2. Enter "user@abc" (no .com/.my).* *3. Click Register.*	No console errors. API returned 200 OK and successfully created a record for an invalid email.	*Missing frontend/backend validation; the system lacked a Regex check to filter for valid email domains before sending data to Supabase.*	*https://github.com/deepnan-umhack/scada-i-umhack26/commit/c17a3082e478b13be16e162a8912c1df943a43ca]*
DEF-03	*test_precool_start_falls_back_to_20_minutes_when_weather_unavailable computes incorrect expected fallback pre-cool time (14:40) instead of expected 13:40 for a 14:00 booking start.*	*1. Run pytest hvac_agent/tests/unit_tests/calculate_weather_aware_precool_start_unittest.py* *2. Observe failure in fallback test asserting startswith("2026-04-24T14:40") while actual is 2026-04-24T13:40:00+08:00*	FAILED hvac_agent\tests\unit_tests\calculate_weather_aware_precool_start_unittest.py::test_precool_start_falls_back_to_20_minutes_when_weather_unavailable - AssertionError: assert False	*1. Application logic correctly subtracts 20 minutes from booking_start.* *2. Test expected value is created with .replace(minute=40) instead of subtracting timedelta(minutes=20), causing incorrect expected hour.*	*https://github.com/deepnan-umhack/scada-i-umhack26/commit/a1d61582a4ff12a1af33633909bacd9b49b83f57*
DEF-04	*REGENERATE does not the thinking process*	*1. Chat with AI any input* *2. When have the output click*	No console errors. UI successfully executes navigation to '/'	*setMessages (removing the old agent bubble) and*	*https://github.com/deepnan-umhack/scada-i-umhack26/com*

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Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
		REGENERATE, the thinking process disappear	instead of updating local state.	*sendToAI are called back-to-back, but React batches state updates asynchronously . This causes a race condition where isLoading and the new empty agent bubble aren't set before the UI re-renders, so the thinking/chat bubble never appears.*	*mit/f4d18e17fbcf407116a8883c84ea91cec08b7bbf*
DEF-05	*Markdown tables are currently not rendering as formatted tables in the UI. Instead, the raw Markdown syntax (pipes and dashes) is displayed as plain text.*	*1. Chat with AI any input and mentioned to output in table 2. The output shows as the markdown format instead of pretty table UI.*	*No console errors; The parser treats table syntax as plain text without the GFM extension.*	*Missing remark-gfm import/plugin. Standard CommonMark does not support tables; the GFM extension is required for grid rendering.*	*https://github.com/deepnan-umhack/scada-i-umhack26/commit/d5f21fe05fc12cc399d1f35edd298d2504eb5ac9*

4. Known Issues & Deferred Technical Debt

It is normal to ship with minor bugs. Document the bugs you found but did not have time to fix. You are highly encouraged to convert these into open GitHub Issues to demonstrate proper defect tracking.

Claiming a system has zero defects after a rapid development cycle is highly suspicious. This section

evaluates your transparency and risk awareness. Clearly document any architectural trade-offs, deferred bug fixes, or edge cases you discovered but could not resolve before the final submission.

Note: Proactively documenting minor issues demonstrates engineering maturity and will not negatively impact your score. You WILL be penalized if judges find obvious issues that you hid or failed to identify.

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
DEF-01	***Technical Debt:*** *No validation on equipment quantity requests. The system successfully processes a booking request that includes unrealistic equipment quantities (e.g., Chair x179, Microphone x200, Camera x100, Projector x100) without validating whether these quantities are logical, available, or necessary for a 4-person meeting. This indicates that the Booking Agent does not enforce constraints between room capacity, actual inventory limits, and requested equipment quantities. While the booking flow completes successfully, this may lead to resource over-allocation, inaccurate inventory usage, and unrealistic system outputs, reducing system reliability.*	*Deferred due to time constraints and prioritization of core end-to-end booking, ESG compliance, and HVAC automation flows. Proper validation requires integrating inventory constraints, capacity-based rules, and possibly database checks, which were not fully implemented in this prototype phase.*	*https://github.com/deepnan-umhack/scada-i-umhack26/issues/55#issue-4369473498*
DEF-02	***UI Glitch:*** *Regenerate button causes repeated text duplication (response bloating). When the user repeatedly clicks the "Regenerate" button, the system appends or re-renders responses without clearing previous outputs, resulting in a large duplicated text block ("blob"). This degrades user experience, reduces readability, and may impact frontend performance or*	*Deferred due to focus on core agent functionality (booking, HVAC, ESG). Fix requires frontend state management improvements (e.g., replacing instead of appending responses, debouncing button clicks, or resetting message state), which was not prioritized during the prototype phase.*	*https://github.com/deepnan-umhack/scada-i-umhack26/issues/56#issue-4369483132*

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Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
	memory usage over time. Although the core functionality remains intact, the UI behavior is not properly controlled.		
DEF-03	***UI Glitch:*** *Booking status text layout breaks in tablet view (responsive UI issue). When viewed in tablet-sized resolution on a laptop, the booking status card displays text in a vertically squashed and misaligned format. Key details such as date, time, room, and equipment are compressed into a narrow column, reducing readability and causing poor visual presentation. While the booking data is still correct, the UI responsiveness is not properly optimized for mid-sized screens, leading to a degraded user experience.*	*Deferred due to prioritization of core system functionality (booking flow, agent orchestration, ESG, HVAC). Fix requires responsive CSS improvements (e.g., adjusting breakpoints, flex/grid layout tuning, and text wrapping rules), which were not fully refined during the prototype phase.*	*https://github.com/deepnan-umhack/scada-i-umhack26/issues/58#issue-4369494713*

5. Live Demo / UAT Risks

Every prototype has operational limits. Document the fragile parts of your system that judges should know about before they test your application. You are required to identify at least two hard limits or specific operational instructions.

Claiming a prototype has zero risks or boundaries after a rapid development cycle is highly unrealistic. This section evaluates your ability to anticipate failure. Clearly define what your system can and cannot handle so the judging panel evaluates you fairly within those parameters.

Note: Proactively documenting system boundaries protects your demo score and demonstrates proper risk management. You WILL be significantly penalized if you claim, "no risks" and your application crashes or fails unexpectedly during the live demo.

Risk 1: LLM Inference Latency (The "Wait" Period)

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- **Description:** The system may experience LLM inference latency of approximately 5–10 seconds for standard requests and up to 20–30 seconds for complex multi-agent workflows (e.g., Booking + HVAC + ESG combined). Users must wait for the response to fully complete before submitting new inputs, as overlapping requests can cause inconsistent states or delayed responses.

Risk 2: MQTT Simulation Synchronicity

- **Description:** This prototype uses a simulated IoT environment via MQTT. Commands sent to the HVAC agent may take up to 4 seconds to reflect in the UI dashboard due to the polling interval of the simulated sensor nodes. Rapidly toggling settings may cause the UI to briefly show old state data before syncing.

Risk 3: Concurrent Booking Conflicts

- **Description:** While the system uses PostgreSQL state management, the prototype does not currently implement 'Live Locking' on the frontend. If two users attempt to book the same room at the exact same second, the first request will succeed in the database, but the second user may need to manually refresh the page to see the updated availability.

Risk 4: Environment Configuration Sensitivity

- **Description:** Core services require env vars (OPENAI_API_KEY, Supabase creds, MQTT settings). Missing/misconfigured env causes startup/runtime failures, including explicit throw in scheduled function. Therefore, we recommend the judges/panelist to assess using the deployed versions.

Risk 5: Data is simulation-driven, not live BMS/SCADA hardware

- **Description:** The replay pipeline publishes historical sensor records every 4 seconds to MQTT topics where it is controlled by simulation behavior, not real hardware telemetry. Judges should expect demo realism, not production-grade device integration resilience.

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